

Heat transfer

1. Conduction.

Energy Balance for Rate of Thermal Energy $\rightarrow$ for basic prob apply Energy Balance
 All prob will be solved

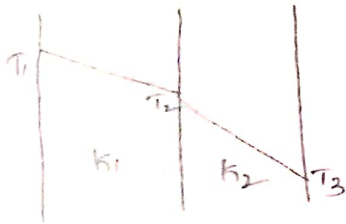
$$Q_{in} + Q_{gen} - Q_{out} = Q_{stored}$$

V.I.M.P.K

Heat conduction equation

steady state $\frac{\partial T}{\partial t} = 0$, $Q_{stored} = 0$

$$\frac{1}{r^n} \frac{\partial}{\partial r} \left(r^n \frac{\partial T}{\partial r} \right) + \frac{q_g}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$



$$\Delta T_1 < \Delta T_2$$

$$R_{th1} < R_{th2}$$

$$k_1 > k_2$$

cartesian $\frac{\partial T}{\partial x} \quad n=0$ $\frac{\partial^2 T}{\partial x^2} + \frac{q_g}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$

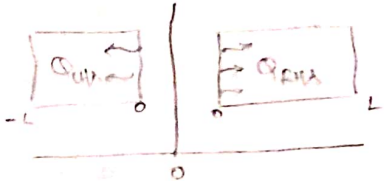
cylindrical $\frac{\partial T}{\partial r} \quad n=1$ $\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{q_g}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$

spherical $\frac{\partial T}{\partial r} \quad n=2$ $\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) + \frac{q_g}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$

Fourier law of Heat conduction

Thermal conductivity (k) W/mK

$$Q_{cond} = kA \frac{\Delta T}{L}$$



when heat flow right hand side

ΔT due to left hand side

$$\frac{\Delta T}{\Delta L} = +ve \quad \frac{\Delta T}{\Delta L} = -ve$$

$$Q_{cond} = -kA \frac{\Delta T}{L}$$

$$Q = kA \frac{\Delta T}{L} = +ve \quad Q = -kA \frac{\Delta T}{L} = -ve$$

$$k_{solid} > k_{liq} > k_{gas}$$

V.I.M.P.K

$$Temp \uparrow \quad k_{non-metal} \uparrow \quad k_{metal} \downarrow \quad k_{liq} \downarrow \quad k_{gas} \uparrow$$

$$\alpha = \frac{k}{\rho c}$$

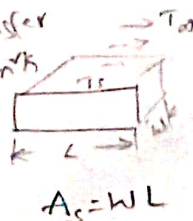
density ρ sp. heat c

$$\text{thermal diffusivity } (\alpha) = \frac{m^2}{s}$$

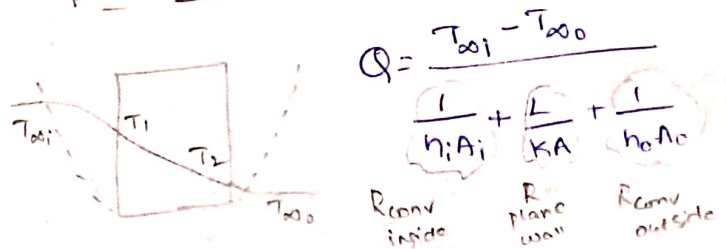
Newton's law of cooling

$$Q_{conv} = h A_s \Delta T$$

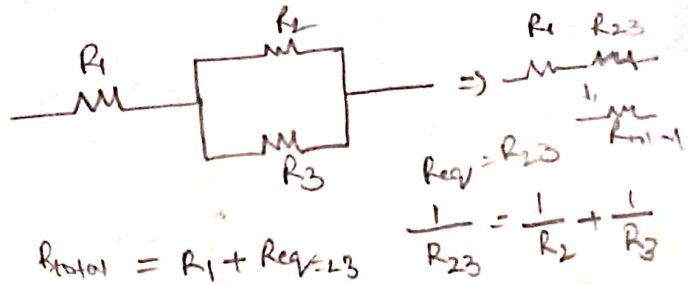
$$Q_{conv} = h A_s (T_s - T_\infty)$$



plane wall



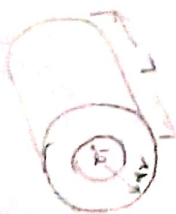
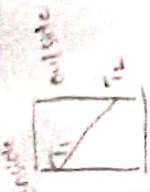
$$Q = \frac{T_{\infty 1} - T_{\infty 2}}{\frac{1}{h_i A_i} + \frac{L}{kA} + \frac{1}{h_o A_o}}$$



$$R_{total} = R_1 + R_{eq} = R_1 + R_2 R_3 / (R_2 + R_3)$$

$$\frac{1}{R_{23}} = \frac{1}{R_2} + \frac{1}{R_3}$$

Geometry



Temp profile

$$\frac{T - T_1}{T_2 - T_1} = \frac{x}{L} = \frac{x - 0}{L - 0}$$

$$\frac{T - T_1}{T_2 - T_1} = \frac{\ln(\frac{r}{r_1})}{\ln(\frac{r_2}{r_1})}$$

$$\frac{T - T_1}{T_2 - T_1} = \frac{\frac{1}{r_1} - \frac{1}{r}}{\frac{1}{r_1} - \frac{1}{r_2}}$$

Linear

Logarithmic

Hyperbolic

Heat Equation

$$Q = \frac{T_1 - T_2}{\frac{L}{KA}}$$

$$Q = \frac{T_1 - T_2}{\frac{\ln(\frac{r_2}{r_1})}{2\pi KL}}$$

$$Q = \frac{T_1 - T_2}{\frac{1}{4\pi K} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)}$$

Formula in terms of Area with comp plane wall

$$Q = \frac{T_1 - T_2}{\frac{\ln(\frac{r_2}{r_1})}{2\pi KL}}$$

$$Q = \frac{T_1 - T_2}{\frac{1}{4\pi K} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)}$$

$$A_{lm} = A_m = A$$

$$A_{lm} = \frac{A_1 - A_2}{\ln(\frac{A_1}{A_2})}$$

$$A_{lm} = A_m = A$$

$$A_{lm} = \sqrt{A_1 A_2}$$

$$A_c = 4\pi r^2$$

Area

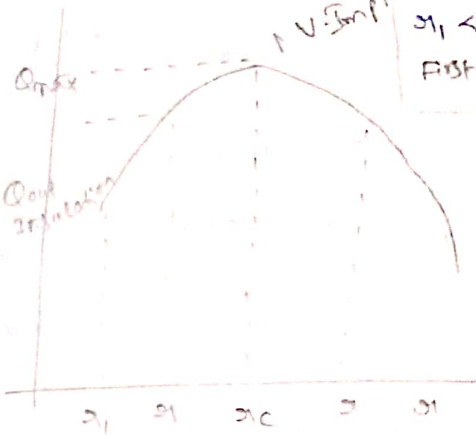
Critical thickness

$$\text{cylinder} = r_c = \frac{k_{ins}}{h_o}$$

$$\text{Sphere} = r_c = \frac{2k_{ins}}{h_o}$$

Insulation thickness = r_c

Insulation thickness = r_c



Internal Temp $T_1 > T_c$ Q always $\uparrow$
 $T_1 < T_c$ First $\uparrow$ then $\downarrow$
 First increase then decrease

$T_1 < T_c$ Initially heat loss increases, reaches to max value then start decreasing

max heat loss $r = r_c$

Imp point

Electric wire $Q_{ins} \geq Q_{without ins}$

Steam pipe $Q_{with ins} < Q_{without ins}$

Insulation Thickness $\uparrow A_c \uparrow Q_{conv} \uparrow Q_{cond} \downarrow$

• Addition of Insulation inside Surf of pipe always red HT rate and critical thickness Concept No Significant

• Insulation inside

Surf resistance $\uparrow$ Internal resistance $\uparrow$

Variable thermal conductivity

$$K = K_0 [1 \pm B T_{avg}]$$

2. Conduction with Internal heat generation

Case 1: Both surf of plane wall main at diff temp

$$T = \frac{-q_g x^2}{2k} + \left(\frac{T_2 - T_1}{L} + \frac{q_g L}{2k} \right) x + T_1$$

→ parabolic Eqn

$$Q_{in} + Q_{gen} - Q_{out} = Q_{stored} \rightarrow \text{Energy Balance Eqn}$$

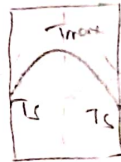
$$Q_{in} = -kA \frac{\partial T}{\partial x} \Big|_{x=0} \quad (\text{or}) \quad hA_s(T_s - T_{\infty})$$

$$Q_{out} = -kA \frac{\partial T}{\partial x} \Big|_{x=L} \quad (\text{or}) \quad hA_s(T_s - T_{\infty})$$

$$Q_{gen} = q_g \times \text{Volume} \quad (\text{or}) \quad \int_0^L q_g A dx$$

$$Q_{stored} = mc \frac{\partial T}{\partial t} \quad m = \rho V$$

Imp



procedure for max prob

$Q_{stored} = 0$

$$\frac{1}{\rho c} \frac{\partial}{\partial t} \left(\rho \frac{\partial T}{\partial x} \right) + \frac{q_g}{k} = \frac{1}{k} \frac{\partial T}{\partial t}$$

plane wall $x=0$ to $x=L$
 $\frac{\partial^2 T}{\partial x^2} + \frac{q_g}{k} = 0$

$$\int \frac{dT}{dx} = - \int \frac{q_g}{k} dx$$

$$\frac{dT}{dx} = - \frac{q_g}{k} x + C_1 \Rightarrow \int \frac{dT}{dx} = - \int \frac{q_g}{k} x + C_1$$

$$T = \frac{-q_g x^2}{2k} + C_1 x + C_2$$



At $x=0$ $T=T_1$ $C_2 = \dots$

$x=L$ $T=T_2$ $C_1 = \dots$

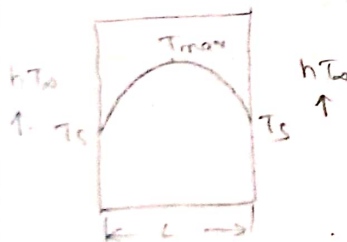
$\frac{dT}{dx} = 0$ (at $4 \times 10^5 x = 200 \Rightarrow x = 5 \times 10^2 = 5 \text{ mm}$ from left side)

$$T_{max} - T_s = \frac{q_g L^2}{2k}$$

T Eqn in Terms of x (eg: $-2 \times 10^5 x^2 + 200x + 160$)

For max Temp $\frac{dT}{dx} = 0$ at x location $T = T_{max}$

plane wall



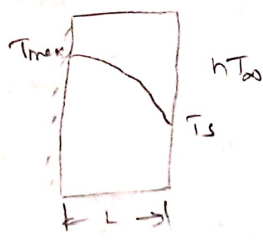
$$T_{max} - T_s = \frac{q_g L^2}{8k}$$

$$T_s - T_{\infty} = \frac{q_g}{h} \times \frac{L}{2}$$

$$\frac{T_{max} - T_{\infty}}{T_{max} - T_s} = 1 + \frac{4}{Bi}$$

(or) $\frac{T_{\infty} - T_{max}}{T_s - T_{max}} = Bi = \frac{hL}{k}$

plane wall



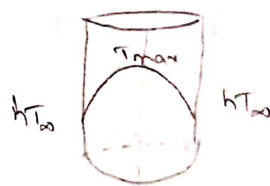
$$T_{max} - T_s = \frac{q_g L^2}{2k}$$

$$T_s - T_{\infty} = \frac{q_g}{h} \cdot L$$

$$\frac{T_{max} - T_{\infty}}{T_{max} - T_s} = 1 + \frac{2}{Bi}$$

(or) $Bi = \frac{hL}{k}$

Solid Cylinder



$$T_{max} - T_s = \frac{q_g R^2}{4k}$$

$$T_s - T_{\infty} = \frac{q_g}{h} \cdot \frac{R}{2}$$

$$\frac{T_{max} - T_{\infty}}{T_{max} - T_s} = 1 + \frac{8}{Bi}$$

(or) $Bi = \frac{hR}{k}$

Solid Sphere



$$T_{max} - T_s = \frac{q_g R^2}{6k}$$

$$T_s - T_{\infty} = \frac{q_g}{h} \cdot \frac{R}{3}$$

$$\frac{T_{max} - T_{\infty}}{T_{max} - T_s} = 1 + \frac{2}{Bi}$$

(or) $Bi = \frac{hR}{k}$

Unsteady (or) Transient Heat Conduction

Biot Num $Bi = \frac{hL_c}{k}$ characteristic len

Fourier Num $f = \frac{\text{rate of heat cond in given Volume}}{\text{rate of thermal energy stored in given Volume}}$

Fourier Num is dimensionless Num or dimensionless time

$$\frac{T - T_\infty}{T_0 - T_\infty} = e^{-\frac{hA_s t}{mc}} = e^{-\frac{hA_s t}{\rho V c}} = e^{-\frac{ht}{\rho L_c c}} = e^{-\frac{t}{\tau^*}} = e^{-Bi \cdot Fo}$$

$$Bi = \frac{hL_c}{k}$$

$$Fo = \frac{\alpha t}{L_c^2}$$

Thermal time Const ($t = \tau^*$)

$$T - T_\infty = 36.78 \% (T_0 - T_\infty)$$

$$T_0 - T = 63.4 \% (T_0 - T_\infty)$$

Whole chapter is depend on this formula only

$$\frac{T - T_\infty}{T_0 - T_\infty} = e^{-\frac{t}{\tau^*}}$$

temp at time t
V.Imp
time
thermal time Const
Sp heat char len

$$\tau^* = \frac{\rho L_c c}{h}$$

$$\frac{T_0 - T}{T_0 - T_\infty} = 1 - e^{-\frac{t}{\tau^*}}$$

$$L_c = \frac{\text{Volume}}{\text{Surf Area}} = \frac{V}{A_s}$$

$$L_c = \frac{R}{3} \text{ Sphere}$$

$$L_c = \frac{R}{2} \text{ Cylinder}$$

$$L_c = \frac{L}{2} \text{ thin plate}$$

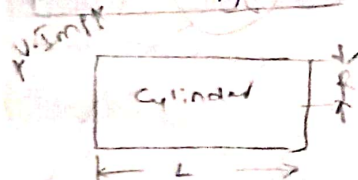
check

$$Bi = \frac{hL_c}{k} < 0.1$$

Lump Body
Biot No minimum

$$R_{cond} \ll R_{conv}$$

$$\frac{\partial T}{\partial t} \text{ negligible}$$



$$V = \pi R^2 L$$

$$A_s = \pi R^2 + 2\pi R L + \pi R^2$$

V.Imp	Volume	Surf Area
Sphere	$\frac{4}{3}\pi R^3$	$4\pi R^2$
Cylinder	$\pi R^2 L$	$\pi R^2 + 2\pi R L + \pi R^2$

Heat loss by Ball = Heat gained by fluid

Heat loss

$$m c_p \frac{\partial T}{\partial t} = h A_s (T_0 - T_\infty)$$

mass of Body

$$m = \rho V$$

$$\alpha = \frac{k}{\rho c}$$

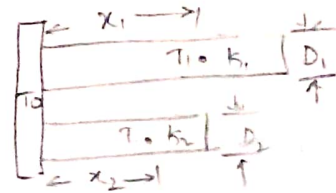
Like that

Fins (Questions are not asking very much from fin in gate)

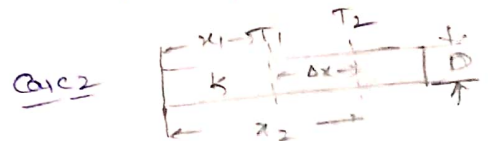
Fischer

Special Case

Case 1: Same base temp diff type of fin



$$\frac{k_2}{k_1} \times \frac{D_2}{D_1} = \left(\frac{x_2}{x_1}\right)^2$$

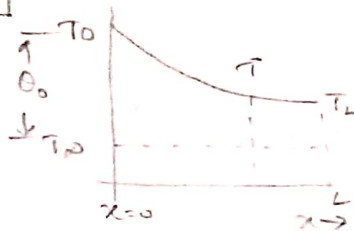


$$\frac{T_1 - T_{\infty}}{T_2 - T_{\infty}} = e^{m \Delta x} \quad \Delta x = x_2 - x_1$$

Infinite length

$$\frac{T - T_{\infty}}{T_0 - T_{\infty}} = e^{-mx}$$

$$m = \sqrt{\frac{hp}{kAc}}$$



$$Q_{fin} = \sqrt{hp k Ac} \times \theta_0$$

Finite length with Insulated tip

$$Q_{fin} = \sqrt{hp k Ac} \times \theta_0 \times \tanh mL$$

$$\theta_0 = T_0 - T_{\infty}$$

$$\frac{T - T_{\infty}}{T_0 - T_{\infty}} = \frac{1}{\cosh mL}$$

$$\text{Effectiveness of fin } (\epsilon) = \frac{Q_{\text{with fin}}}{Q_{\text{without fin}}}$$

$$\frac{\sqrt{hp k Ac} \times \theta_0 \times \tanh mL}{h A_c \theta_0}$$

$$\epsilon = \frac{\sqrt{kp} \tanh mL}{\sqrt{h A_c}}$$

$$\text{Efficiency of fin } \eta = \frac{Q_{fin}}{Q_{max fin}} = \frac{\sqrt{hp k Ac} \times \theta_0}{h P L \theta_0}$$

$$\eta_{\text{infinite length}} = \frac{1}{mL}$$

$$\eta_{\text{finite length}} = \frac{\tanh mL}{mL}$$

$$\frac{\epsilon}{\eta} = \frac{\text{Surf area of fin}}{\text{Cross section area of fin}} = 4 \left(\frac{L}{D}\right)$$

$$\frac{h}{mk} = 1 \quad Bi = 1 \quad \text{No use of fin}$$

$$\frac{h}{mk} > 1 \quad Bi > 1 \quad \text{Fin act as Heat Insulator (No at all recommended)}$$

$$\frac{h}{mk} < 1 \quad Bi < 1 \quad \text{Fin act as Heat Conductor (Best for HT transfer)}$$

Convection

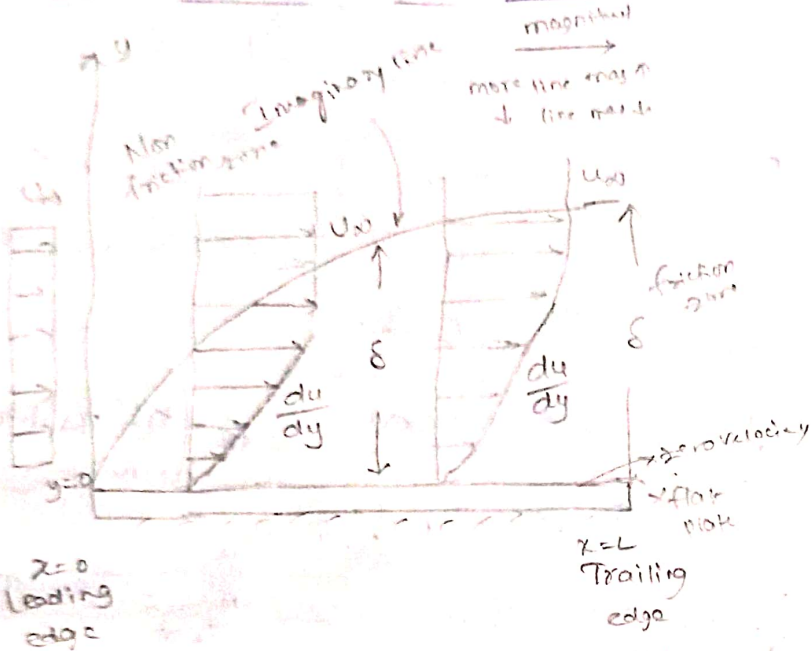
(Zusammenlagerung)

(fluid motion)

Forced Conv. External flow

$$h_{\text{boiling (or) condensation}} > h_{\text{forced conv liq}} > h_{\text{forced conv gas}} > h_{\text{free conv liq}} > h_{\text{free conv gas}}$$

Hydrodynamic (Velocity) Boundary layer



δ = Hydrodynamic Boundary Layer thickness

$$\frac{du}{dy} = \text{Velocity gradient}$$

$$\text{Reynolds Num (Re)} = \frac{\text{Inertia force}}{\text{Viscous force}} = \frac{\rho U_{\infty} L \text{ (or } D)}{\mu} = \frac{\rho U_{\infty} L}{\mu}$$

Thermal diffusivity (α) = $\frac{k}{\rho C}$ $\frac{m^2}{s}$

$$\text{Prandtl Num (Pr)} = \frac{\text{momentum diffusion}}{\text{Thermal diffusion}}$$

$$P_g = \frac{U}{\alpha} = \frac{\mu E_p}{K}$$

$$P_r = \left(\frac{\delta}{\delta T} \right)^3$$

If $P_0 = 1$ $\delta = \delta_T$

$$P_Y > 1 \quad \delta > \delta_T$$
$$p_1 < 1 \quad \delta \ll \delta_T$$

Laminar
flow

Turbulent flow

$$S = S_T \quad (\text{Due to Intensive mixing})$$

$$(Pr)^{1/3} = \frac{\delta}{\delta_T}$$

* V.V Imp *

- Signal ~~term~~
gradient

By applying this cell max
all problems solved in
the fixed conv external flow

8V.3m9

5012

Handwritten: *Handwritten*



Reynold's
(Re) Num

Flow

hydrodynamic
(S) thickness

$$h \propto \frac{1}{\delta T}$$

Skin-friction coefficient (C_f)

$$Re \leq 5 \times 10^5$$

Lamina 1

$$\frac{0.664}{\sqrt{Re_x}}$$

$$R_e > 10.4$$

Turbulent

0.3582
1/5
Res

$$\frac{0.059}{115}$$

Turbulent flow over flat plate ($T_s = \text{const}$)

$$Nu_x = 0.029 Re_x^{4/5} Pr^{1/2}$$

max prob procedure for flat plate conv prob

(recharge in seq)

$$T = 304 + 257^2 + 67^3 (T_s - T_{\infty}) + 100$$

max prob
it is air

$$= V_{cond} \cdot C \cdot V$$

$$\frac{dT}{T_{\text{film}}} = h(T_s - T_{\text{fluid}})$$

Forced Convection Internal flow

$$U_m = \frac{2}{R^2} \int_0^R U(r,x) r dr$$

$$T_m = \frac{2}{R^2 U_m} \int_0^R U(r,x) T(r,x) r dr$$

For Laminar $P_r = \frac{L_T}{L_h}$
 thermal entry length L_T
 hydrodynamic entry length L_h

Turbulent $L_T = L_h$

$Re < 2300$ Laminar flow
 $Re > 10,000$ Turbulent flow

$$dQ = m \dot{C}_p dT_m = q \cdot P \cdot dx$$

$$Q = \int_{T_i}^{T_o} m \dot{C}_p dT_m = \int_0^L q P dx$$

$$Q = m \dot{C}_p (T_o - T_i) = q P L$$

Energy Balance Eqn

P = perimeter

m = mass flow rate

$$P = \pi D$$

$$P = 2(a+b)$$

$$m = \rho \times U_{avg} \times A_c$$

density ρ , velocity U_{avg} , cross section area A_c

$$Q = U_{avg} \times A_c \quad m^3/s$$

Volume flow rate or discharge

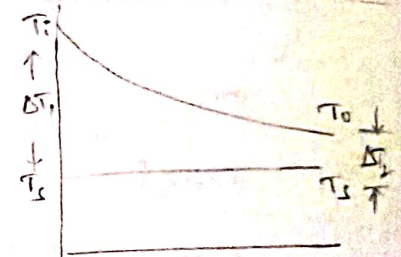
Apply this Eqn most of prob are solved.

$$q = h(T_s - T_o)$$

$$\frac{q}{h} = (T_s - T_o)$$

$$(T_s - T_o) = \text{const}$$

In thermal fully dev



$$Q = h A_s \text{ LMTD}$$

$$Nu = \frac{h D_h}{k}$$

$$Re = \frac{U_{avg} D_h}{\mu}$$

$$D_h = \frac{4 A_c}{P}$$

A_c = cross section area, P = perimeter

$$\text{LMTD} = \frac{\Delta T_1 - \Delta T_2}{\ln \left(\frac{\Delta T_1}{\Delta T_2} \right)}$$

At $T_s = \text{const}$ B.C. $Nu = 3.66$

$$Nu = \frac{h D}{k} = 3.66 \quad Re < 2300$$

At $q = \text{const}$ B.C. $Nu = 4.36$

$$Nu = \frac{h D}{k} = 4.36 \quad Re < 2300$$

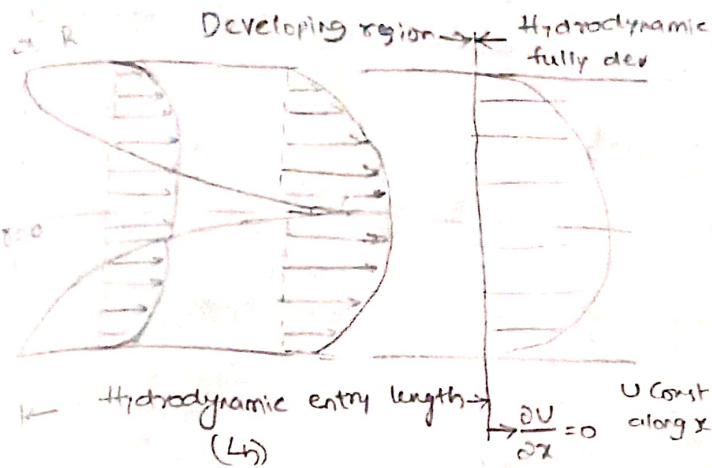
Laminar flow

$$Nu_{q,c} > Nu_{T,c}$$

Shape of Temp profile changes w.r to location $T_s = \text{const}$

Shape of Temp profile not changes w.r to location $q = \text{const}$

Hydrodynamic (velocity) Boundary Layer



No slip
at $x=0, r=R$

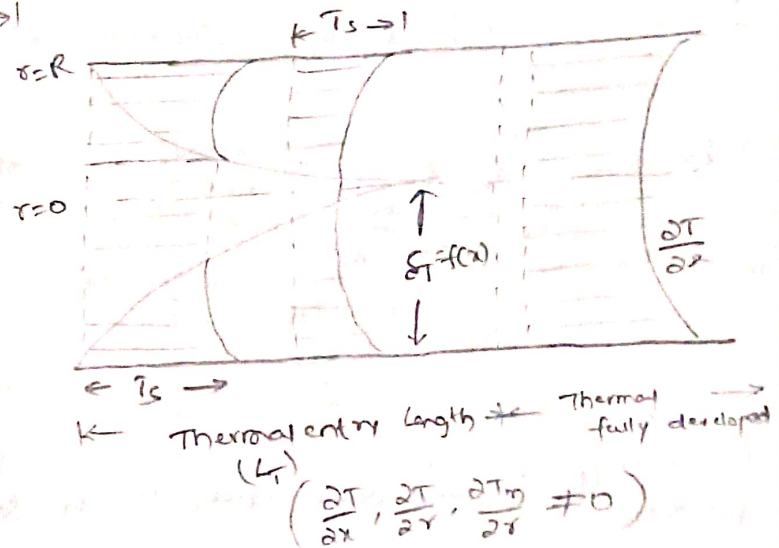
$\frac{\partial U}{\partial x} = 0$
U const along x
entry in fully developed

Laminar flow $L_h = 0.05 Re D$ $Re \uparrow L_h \uparrow$

Turbulent flow $L_h = 10D$

Thermal Boundary Layer

$T_c \gg T_{\infty}$



Laminar flow $L_h = 0.05 Re D \cdot Pr$

Turbulent flow $L_t = 10D$ ($L_t = L_h$)

free convection

$T_c = \text{const}$ $Nu = C [Gr \cdot Pr]^n$

$n = \frac{1}{4}$ Laminar flow $C \& n = \text{const } C > 1$

$n = \frac{1}{3}$ Turbulent flow

$q = \text{const}$

$Nu = C [Gr^* \cdot Pr]^n$

$n = \frac{1}{5}$ Laminar flow

$C > 1$

$n = \frac{1}{4}$ Turbulent flow

$Nu \propto (Gr \cdot Pr)^{1/4}$

$\frac{hL}{k} \propto \left[\frac{g \beta \Delta T L_c^3}{\nu^2} \times \frac{\nu}{\alpha} \right]^{1/4}$

over all h-t coeff (U)

$U_i = U_o = \frac{1}{h_i + h_o}$



$Q = \frac{\Delta T}{\frac{1}{h_i A_i} + R_{\text{cond}} + \frac{1}{h_o A_o}} = U_i A_i \Delta T = U_o A_o \Delta T$

✓ Biot Num $Bi = \frac{\text{Internal thermal resistance}}{\text{Boundary layer thermal resistance}} = \frac{R_{\text{cond offered in solid}}}{R_{\text{conv}}}$

✓ Grashof Num $Gr = \frac{\text{Buoyancy force}}{\text{Viscous force}}$

✓ Prandtl Num $Pr = \frac{\text{momentum diffusivity}}{\text{Thermal diffusivity}}$

✓ Reynolds Num $Re = \frac{\text{Inertia force}}{\text{Viscous force}}$

✓ Nusselt Num $Nu = \frac{Q_{\text{conv}}}{Q_{\text{cond}}}$

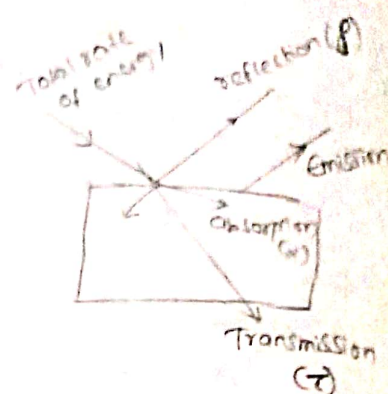
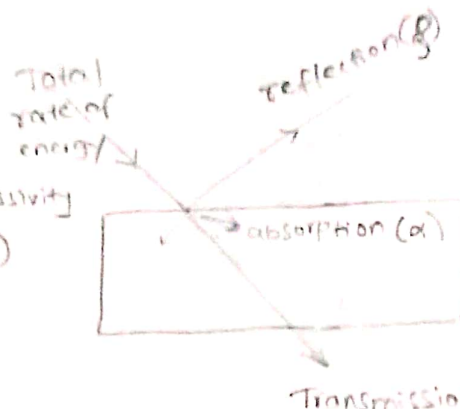
Fourier Num $Fo = \frac{\text{Rate of heat cond in given volume}}{\text{Rate of thermal energy stored in given volume}}$

Radiation

Irradiation (G) Energy reflected + Energy absorbed + Transmitted

$$1 = \rho + \alpha + \tau$$

reflectivity absorptivity Transmissivity
(Solid = 0)



Emissive power (E): How much energy is transferred due to temp

$$\text{Emissivity } (\epsilon) = \frac{E_{\text{any body}}}{E_{\text{Black Body}}}$$

$\times V \cdot \text{Imp}^{\circ} K$

Radiosity (J)

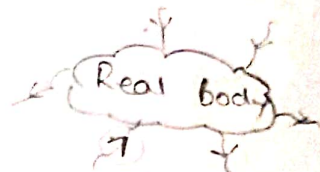
$$J = \rho G + E + \tau G$$



Only absorption

$$\rho = 0 \quad \alpha = 1, \epsilon = 1$$

$\times V \cdot \text{Imp}^{\circ} K$



Reflection present

$$\alpha < 1, \epsilon < 1$$

Wien's displacement law

$$\lambda_1 T_1 = \lambda_2 T_2 = \lambda_3 T_3 = c$$

Temp Wave length

radiation const

Kirchoff's law $\alpha = \epsilon$ At thermal equilibrium (Source = receiver temp)

But in reality if diff in hundreds also proceed.

Stefan Boltzmann law

$$E_b = \sigma \times T^4$$

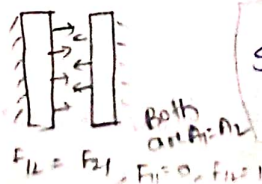
Total emissive power of Black Body

$\times V \cdot \text{Imp}^{\circ} K$

$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \text{K}^4$

Shape factor (or) View factor (or) Geometry factor (or) Configuration factor

F_{ij} emitter receiver
 F_{ji} emitter receiver



Surf area

Sphere	Cylinder
$4\pi r^2$	$\frac{\pi}{4} d^2 + \pi d l + \frac{\pi}{4} d^2$

Circle Surf area = πr^2

Reciprocity rule

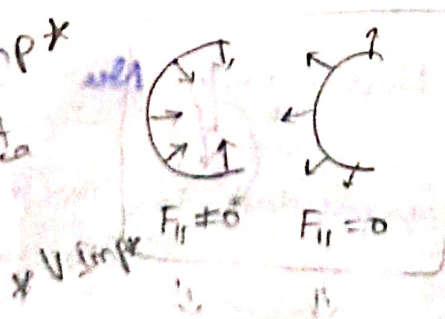
$$A_1 F_{12} = A_2 F_{21}$$

Surf area

$$F_{11} + F_{12} = 1$$

$$F_{22} + F_{21} = 1$$

$\times V \cdot \text{Imp}^{\circ} K$

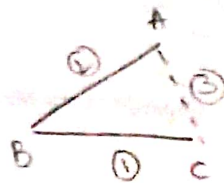


Here radiation is touching the sides also in fig (1) so it is not zero

$$F_{12} = \frac{L_1 + L_2 - L_3}{2L_1}$$

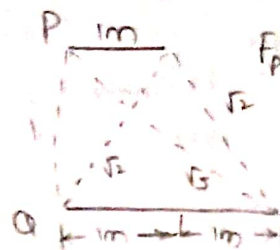
$$F_{13} = \frac{L_1 + L_3 - L_2}{2L_1}$$

$$F_{23} = \frac{L_2 + L_3 - L_1}{2L_2}$$



$$F_{21} + F_{22} + F_{23} = 1$$

Hottle string method

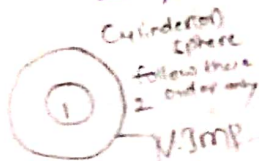


$$F_{PQ} = \frac{(\text{Sum of Cross strings}) - (\text{Sum of Uncross strings})}{2 \times \text{Length of emitter}}$$

$$\frac{(\sqrt{2} + \sqrt{5}) - (1 + \sqrt{2})}{2 \times 1}$$

Net Heat exchange b/w two bodies

Both bodies are grey



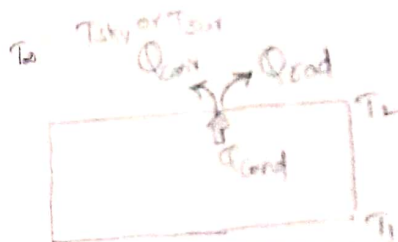
$$Q_{\text{Net}} = \frac{A_1 \sigma (T_1^4 - T_2^4)}{\left(\frac{1}{\epsilon_1} - 1\right) + \frac{1}{F_{12}} + \frac{A_1}{A_2} \left(\frac{1}{\epsilon_2} - 1\right)}$$

prob solved by these single formula only

Radiation

$$Q = \epsilon \sigma A (T_1^4 - T_{\text{sur}}^4)$$

$T_{\text{sky}} = T_{\text{sur}}$ If T_{sky} not given take T_{a}

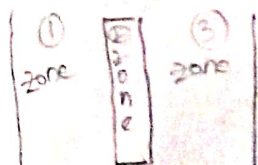


At study state $Q_{\text{in}} = Q_{\text{out}}$

$$\left[\frac{T_1 - T_2}{L} \right] = h (T_2 - T_{\infty}) + \epsilon \sigma (T_2^4 - T_{\text{sur}}^4)$$

$$Q_{\text{cond}} = Q_{\text{conv}} + Q_{\text{rad}}$$

At Study state $Q_1 = Q_2 = Q_3$



Under study state

$$Q_{\text{in}} = Q_{\text{out}}$$

$$\alpha_{\text{sol}} G A = \epsilon \sigma (T_s^4 - T_{\infty}^4) \times A$$

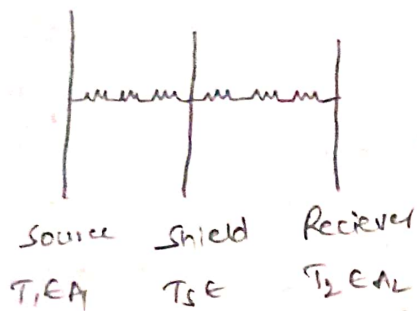
Small Body kept in large ambient

$\alpha = 0.4$ $T_{\infty} = 25^\circ\text{C} = 298\text{K}$



$\alpha = \epsilon$ According to Kirchhoff's law

Radiation with shield (this topic only once asked in gate not asking very much)



$$Q_{\text{without shield}} = \frac{A\sigma(T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1}$$

$$Q_{\text{with shield}} = \frac{A_1\sigma(T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} + \frac{2N}{\epsilon_3} - (N+1)}$$

$$\frac{Q_{\text{with shield}}}{Q_{\text{without shield}}} = \frac{1}{N+1}$$

only remember this formula.
No of shields

$$\frac{T_1^4 - T_s^4}{T_s^4 - T_2^4} = \frac{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_3} - 1}{\frac{1}{\epsilon_3} + \frac{1}{\epsilon_2} - 1}$$

$$\% \text{ red in H.T} = \frac{Q_{\text{without shield}} - Q_{\text{with shield}}}{Q_{\text{without shield}}}$$

Radiation effect on temp measurement (not much asking in gate) Skip this topic

$$Q_{\text{cond}} = Q_{\text{rad}}$$

$$h(T_f - T_{th}) = \epsilon\sigma(T_{th}^4 - T_w^4)$$

$$T_f = T_{th} + \frac{\epsilon\sigma}{h}(T_{th}^4 - T_w^4)$$

Actual temp of fluid

Thermometer temp

Temp of surrounding

Heat Exchangers

Q. Imp

$$Q = \dot{m}_h c_h (T_{h1} - T_{h2}) = \dot{m}_c c_c (T_{c2} - T_{c1}) = UA_s \text{ LMTD}$$

$$\text{LMTD} = \frac{\Delta T_1 - \Delta T_2}{\ln \left(\frac{\Delta T_1}{\Delta T_2} \right)}$$

able lesson run on this formula

$C_{\min} \rightarrow \Delta T \uparrow$ at given instance of time
Heat capacity rate

$$C_h (T_{h1} - T_{h2}) = C_c (T_{c2} - T_{c1})$$

if $C_h < C_c$, $(T_{h1} - T_{h2}) > (T_{c2} - T_{c1})$

$C_c < C_h$, $(T_{c2} - T_{c1}) > (T_{h1} - T_{h2})$

$C_h = C_c$, $(T_{h1} - T_{h2}) = (T_{c2} - T_{c1})$

C Capital C
Heat Capacity
c Small c
Sp. heat

F = Correction factor (f cal from graph)

$$F = \frac{\text{LMTD}_{\text{cross or multipass}}}{\text{LMTD}_{\text{counter}}}$$

F=1 No correction

Heat capacity ratio (R)

$$R = \frac{C_{\min}}{C_{\max}} \quad R \leq 1$$

For Balanced Heat exchanger (H.E)

$$C_h = C_c \quad R=1 \quad \text{V Imp}$$

$$\text{LMTD}_{\text{counter}} > \text{LMTD}_{\text{cross}} > \text{LMTD}_{\text{parallel}} \quad \text{V Imp}$$

$$\text{Area}_{\text{counter}} < \text{Area}_{\text{cross}} < \text{Area}_{\text{parallel}}$$

For phase change devices
Boiler or condenser

$$C_{\max} \rightarrow \infty \quad R=0 \quad \text{V Imp}$$

Special Case

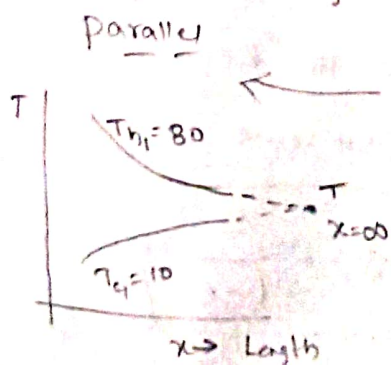
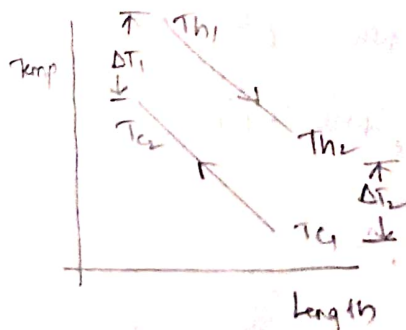
$C_h = C_c$ in counter flow H.E (or) Balanced H.E

$$\Delta T_1 = \Delta T_2$$

$$\Delta T_1 < \Delta T_2 \quad \text{LMTD} < \text{AMTD}$$

$$\Delta T_1 > \Delta T_2 \quad \text{LMTD} < \text{AMTD}$$

$$\Delta T_1 = \Delta T_2 \quad \text{LMTD} = \text{AMTD}$$



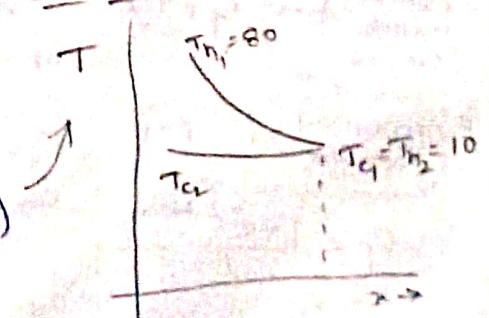
$$\dot{m}_h c_h (T_{h1} - T) = \dot{m}_c c_c (T_c - T_{c1})$$

$$1(80 - T) = 2(T - 10)$$

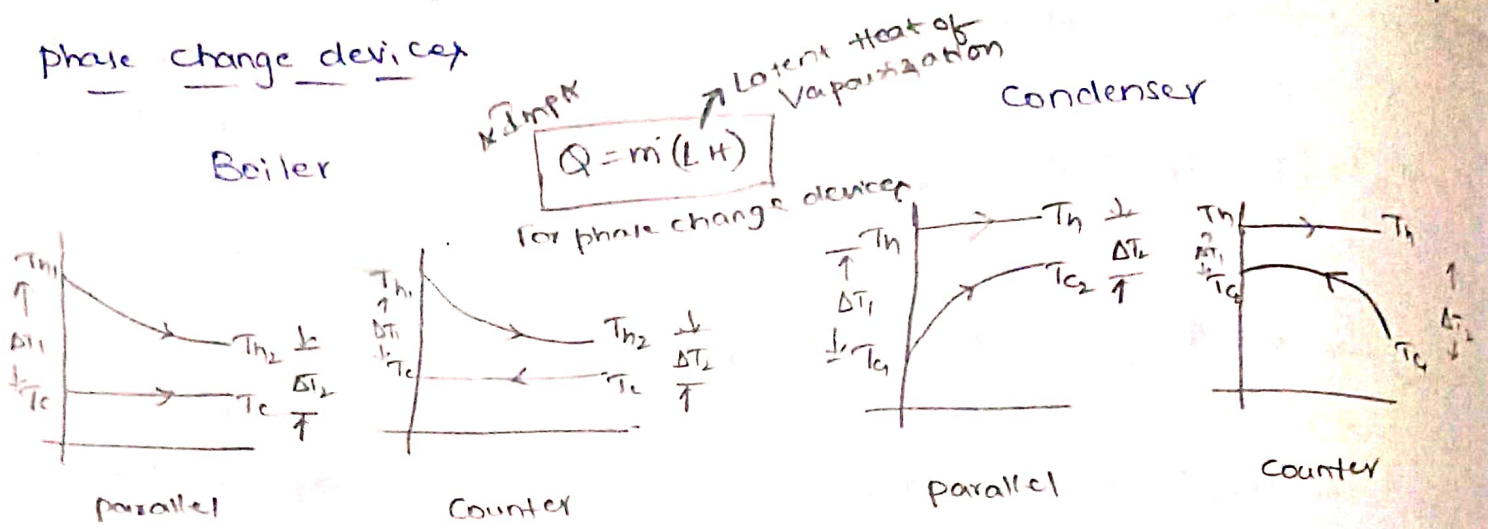
$$\dot{m}_h c_h (T_{h1} - T_{h2}) = \dot{m}_c c_c (T_{c2} - T_{c1})$$

$$1(80 - 10) = 2(T_{c2} - 10)$$

Counter



phase change devices



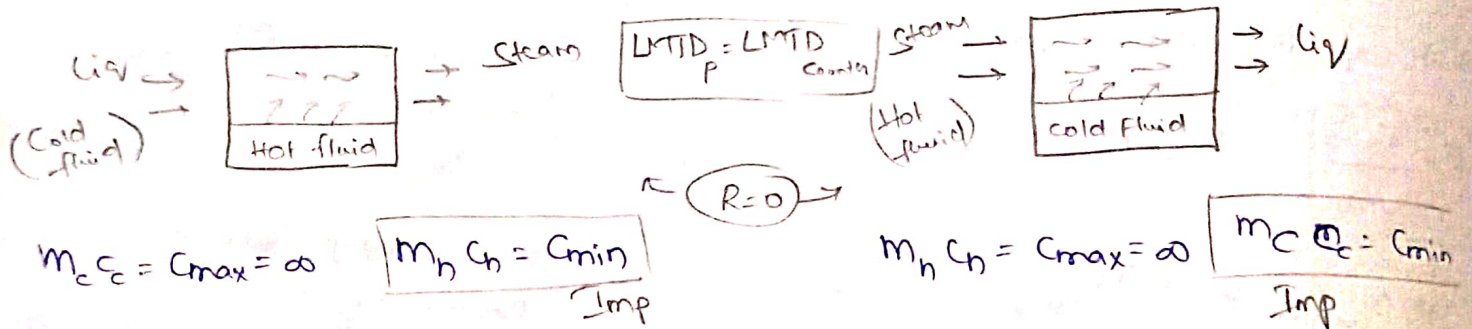
$$Q = m_c h_{fg} = m_h c_h (T_{h1} - T_{h2}) = UA LMTD$$

Latent heat J/kg

$$Q = m_h h_{fg} = m_c c_c (T_{c2} - T_{c1}) = UA LMTD$$

Cold fluid under phase change

Hot fluid under phase change



Effectiveness (ϵ) - NTU method

$$\epsilon = \frac{Q_{act}}{Q_{max}} = \frac{Q_{act}}{C_{min}(T_{h1} - T_{c1})}$$

$$NTU = \frac{UA}{C_{min}} \quad R = \frac{C_{min}}{C_{max}} \quad \epsilon \propto Q_{act} \propto NTU \propto A$$

ϵ for parallel flow

ϵ for counter flow

$$\epsilon_p = \frac{1 - e^{-NTU(1+R)}}{1+R}$$

$$\epsilon_c = \frac{1 - e^{-NTU(1-R)}}{1 - R e^{-NTU(1-R)}}$$

$\therefore$ phase change device $R=0$

i. phase change device $R=0$

$$\epsilon_p = 1 - e^{-NTU}$$

ii. Balanced parallel HE $R=1$

$$\epsilon_p = \frac{1 - e^{-2NTU}}{2}$$

iii. Balanced counter flow HE $R=1$

Heat Capacity

$$C_h = C_c \quad \epsilon_c = \frac{1 - e^{-NTU(1-1)}}{1 - R e^{-NTU(1-1)}} = \frac{0}{0} \quad \text{L'H rule}$$

$$\epsilon_c = \frac{NTU}{1+NTU} \quad \text{or} \quad NTU = \frac{\epsilon}{1-\epsilon}$$